

ADARSH SINGH

Software Engineer — Distributed Systems & Applied AI

Noida, Uttar Pradesh, India

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GitHub | GitLab | LeetCode | CodeChef | Portfolio | Open to Relocation

Professional Summary

Backend engineer specializing in distributed systems and autonomous AI agent infrastructure. Shipped a production trading engine processing **10,000+** messages/sec at sub-**5ms** latency, an autonomous multi-agent system (**Rust**) that resolves GitHub issues end-to-end with a **100%** benchmark resolve rate, and a WASM-sandboxed AI tool server sustaining **600 req/s at 0% errors** under load. **1868-rated** competitive programmer (**600+** problems solved).

Education

JSS Academy of Technical Education (AKTU)

Oct 2023 – Oct 2027

Bachelor of Technology in Information Technology (Current CGPA: 7.56)

Noida, India

Work Experience

HELLO LIVE NA Pvt. Ltd.

May 2026 – July 2026

Software Development Engineer Intern (Backend & AI)

Remote, India

- Architected a high-frequency algorithmic trading engine in **Node.js, Python, and WebSockets**, ingesting real-time tick data from multiple exchange APIs at **10,000+ messages/sec** with sub-**5ms** latency.
- Developed a predictive execution module combining **LLM APIs and time-series forecasting** for live market sentiment analysis, lifting backtested portfolio ROI by **15%**.
- Built a fault-tolerant **PostgreSQL + Redis** backend for state management and order-book caching, cutting database response times by **30%** and sustaining high uptime through market volatility.

Projects

Auroth | Autonomous Multi-Agent GitHub Issue Resolver 🔗

June 2026 – Aug 2026

Rust, Axum, Next.js, PostgreSQL, Redis, E2B

- Architected an async agent-orchestration state machine in **Rust (Axum + Tokio)** that autonomously plans, codes, tests, and opens production GitHub PRs end-to-end, achieving a **100%** resolve rate across a self-curated benchmark suite with automatic 3-attempt retry logic.
- Sandboxed LLM-generated code execution via **E2B** containers and built a multi-provider LLM failover layer (Groq, Gemini, Ollama, Anthropic) to eliminate single-provider downtime risk.
- Deployed a full production stack (Render, Vercel, Supabase, Upstash) with GitHub App webhook automation and a live-polling dashboard for real-time execution-trace visualization.

Isolyth | Sandboxed MCP Tool Server for AI Agents 🔗

June 2026 – July 2026

Python, WASM (wasmtime), Redis, PostgreSQL, Docker

- Built a production-grade **MCP** tool server enabling AI agents to safely execute real-world tools (DB queries, file I/O, web fetches, code exec) via a **WASM-sandboxed (wasmtime)** engine enforcing memory limits, instruction fueling, and wall-clock timeouts.
- Hardened against SSRF and path-traversal, added JWT auth, Redis-backed rate limiting, and OpenTelemetry/Prometheus tracing, plus a real-time React dashboard visualizing live tool-call traces and latency.
- Load-tested with Locust at 100 concurrent users, sustaining **535–600 req/s at 0% errors** — proving WASM sandboxing overhead is minimal (**15–30ms**); shipped Docker Compose + GitHub Actions CI/CD to Railway/Vercel.

Multi-Agent Orchestrator | LangGraph System with Human Governance 🔗

May 2026 – June 2026

Python, LangGraph, PostgreSQL, Redis, Celery, FastAPI, Next.js

- Architected a supervisor-worker multi-agent engine (**LangGraph**) coordinating 4 specialized agents (Research, Analysis, Writer, Human-Review), achieving **100%** autonomous task completion across 5 end-to-end test scenarios.
- Implemented PostgreSQL-backed checkpointing for **100%** crash-recovery across restarts and a human-in-the-loop escalation mechanism (3/3 correct escalations, **0%** false-positive rate).
- Decoupled long-running workflows from the API layer via Redis/Celery async task queuing and eliminated a critical supervisor infinite-loop failure mode using a deterministic short-circuit safeguard, validated across 4 documented failure modes.

Achievements

Competitive Programming: LeetCode Knight, Peak Rating 1868; 600+ algorithmic problems solved across arrays, graphs, and DP; 300-day max streak. Active on CodeChef.

Certifications

Oracle: Cloud Infrastructure 2025 Certified AI Foundations Associate

Oct 2025

Google Cloud: Introduction to Generative AI

Sep 2025

DeepLearning.AI: Building AI Voice Agents for Production

Feb 2025

Technical Skills

Core CS: Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems, Computer Networks

Languages: C++, Go, Rust, JavaScript (ES6+), TypeScript, Python, SQL

Frontend & Architecture: React.js, Next.js 14/15, Tailwind CSS, Turborepo (Micro-frontends), WebAssembly

Backend & APIs: Node.js, Express.js, FastAPI, RESTful APIs, Microservices, WebSockets, Pub/Sub

Databases & Caching: PostgreSQL, MongoDB, Redis, Supabase, Mongoose ODM, Query Optimization

AI / Agentic Systems: OpenAI API, Gemini API, LangGraph, MCP, Multi-Agent Orchestration, Prompt Engineering, LLMOps

DevOps, Auth & Tools: Git, Docker, CI/CD (GitHub Actions), Linux, Vercel, Render, Clerk, JWT